DeltaSense

DeltaSense is an experimental monitoring and predictive maintenance platform for 3D printers built on Raspberry Pi and OctoPrint.

The project started from a simple question:

Is it possible to detect mechanical degradation in a 3D printer before it becomes a failure?

To answer this, DeltaSense measures the **real axis movement time** and compares it with the **theoretical movement time**, calculated from distance and feedrate.

The difference between these two values generates the system's core metric:

Δt (Delta t) = Real Time – Theoretical Time

Starting from this indicator, DeltaSense builds a historical dataset of machine movements and applies lightweight AI and statistical models to detect anomalies and identify degradation trends.

Architecture

The platform is composed of independent microservices:

- **Reader Service** → collects data from OctoPrint APIs
- **Commands Service** → sends diagnostic G-code commands (M105, M114, M119)
- **Serial Parser** → analyzes the OctoPrint serial log
- **Flask Dashboard** → provides monitoring and axis testing interfaces
- **AI Engine** → performs predictive analytics
- **Local Chatbot** → explains machine behavior in natural language

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Features

✅ Real-time axis movement measurements

✅ Real vs. theoretical time comparison

✅ Historical movement database

✅ Per-axis baselines

✅ Health Index

✅ Z-Score anomaly detection

✅ Linear Regression for degradation forecasting

✅ Robust isolation-lite detector

✅ Local web dashboard

✅ Diagnostic chatbot

Objective

The purpose of DeltaSense is not to replace maintenance engineers, but to provide a tool capable of:

- detecting early mechanical changes;
- continuously monitoring machine health;
- supporting predictive maintenance activities;
- creating an objective historical record of printer performance.

Vision

DeltaSense aims to bring concepts commonly found in Industry 4.0—condition monitoring, predictive maintenance, and AI-assisted diagnostics—into the world of 3D printers and small print farms using affordable, open-source hardware.

Core Philosophy

Measure. Understand. Predict.